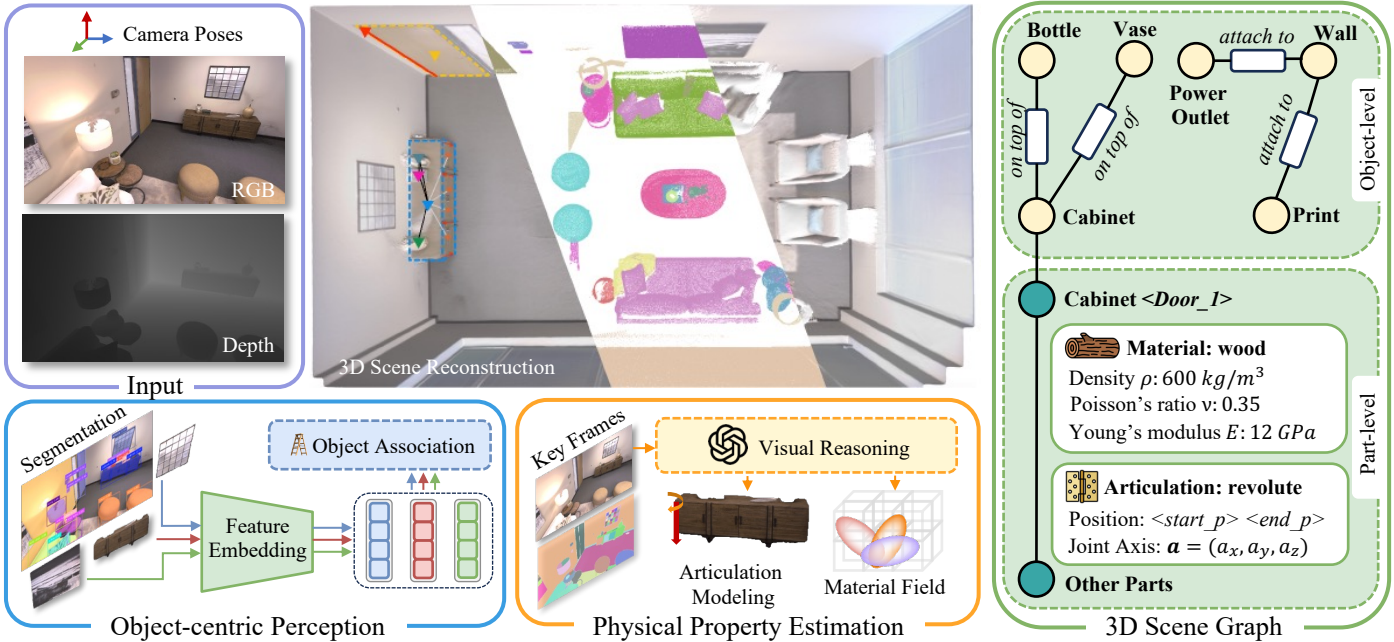
Camera Poses
RGB
Depth
Input
3D Scene Reconstruction
Segmentation
Object Association
Feature Embedding
Object-centric Perception
Key Frames
Visual Reasoning
Articulation Modeling
Material Field
Physical Property Estimation
Bottle
Vase
attach to
Wall
Power Outlet
on top of
on top of
attach to
Cabinet
Print
Object-level
Cabinet <Door_1>
Material: wood
Density ρ: 600 kg/m³
Poisson's ratio ν: 0.35
Young's modulus E: 12 GPa
Articulation: revolute
Position: <start_p> <end_p>
Joint Axis: a = (a_x, a_y, a_z)
Part-level
Other Parts
3D Scene Graph